

KRISHNASAI ADDALA

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[LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [Website](#)

SUMMARY

AI/ML Engineer with experience building production LLM applications, scalable ML infrastructure, and evaluation frameworks across enterprise and research environments. Skilled in GenAI, agentic AI, NLP, MLOps, and explainable AI, with expertise deploying AI systems from research to production.

EXPERIENCE

Self-Employed

AI / ML Engineer - (Model explainability, Opinion dynamic modelling)

Dec 2025 –Present

- Ideated and built AI products centered on agentic systems, LLM evaluation frameworks, and model explainability from concept to implementation.
- Engineered simulation and benchmarking frameworks for opinion dynamics modelling, tool calling, and multi-step reasoning in small language models.

Capgemini

New York City, NY

Associate Consultant — GenAI Developer

Jun 2025 – Nov 2025

- Built and productionized LLM-powered APIs using LangGraph, OpenAI APIs, and NLP techniques, designing end-to-end workflows for retrieval, validation, orchestration, and monitoring to improve regulatory compliance, customer intelligence, and model reliability using MCP integrations for multimodal data ingestion.
- Engineered scalable ML infrastructure on GCP using Airflow, BigQuery, Pub/Sub, and Cloud Functions, developing real-time data pipelines and automating model deployment, versioning, and orchestration across development, staging, and production environments.

Project Geminae — MSBA Capstone (Energy Sector Client)

Pittsburgh, PA

Machine Learning Engineer

Jan 2025 – Apr 2025

- Designed and Built predictive ML models for oil-well selection, leveraging feature engineering, leakage prevention, and model optimization to improve predictive accuracy.
- Applied SHAP explainability to identify key productivity drivers, enabling interpretable predictions and data-driven engineering decisions.

MIDAS Lab, IIIT Delhi

New Delhi, India

Undergraduate Research Engineer

Jun 2023 – May 2024

- Designed and Built a knowledge graph-augmented LLM system for physics question answering, achieving ~200% improvement over baseline and contributing to three published papers.
- Designed and developed LLM evaluation and benchmarking pipelines across retrieval, reasoning, answer quality, and structured outputs, while implementing prompt optimization, LoRA/PEFT fine-tuning, and RLHF-informed evaluation methodologies, driving cross-functional development across retrieval, modeling, and data pipelines.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Java, Bash, ReactJS, HTML

GenAI/Agentic Dev: LangChain, LangGraph, Google ADK, OpenAI API, HuggingFace Transformers, Function Calling, Structured Outputs, Prompt Optimization, AI Agents (Harnesses), RAG, Vector DBs (Chroma, FAISS), Agentic workflows, Tool use, Memory, Planning, LoRA, Fine-Tuning, PEFT, RLHF, MCP, Ollama

ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CUDA, JAX, SHAP, LIME, DiCE, Keras, OpenCV

Cloud/Infrastructure: GCP (BigQuery, Pub/Sub, Cloud Functions, Vertex AI, Airflow), AWS, Docker, Kubernetes, MLflow, SpringBoot, LangSmith

Backend & APIs: FastAPI, Flask, REST APIs, Streamlit, SSE Streaming, Postman

Data: Pandas, NumPy, Spark, Databricks, PostgreSQL, MongoDB, Neo4j, Power BI, Tableau, D3.js

EDUCATION

- Carnegie Mellon University | M.S. in Business Analytics (MSBA) | Pittsburgh, PA | May '25
- Indraprastha Institute of Information Technology Delhi (IIIT-Delhi) | Delhi, India | B.Tech. in Computer Science | May '24

PUBLICATIONS

- Knowledge Graphs are all you need: Leveraging KGs in Physics Question Answering — arXiv, Dec 2024 (arxiv.org/abs/2412.05453)
- Steps are all you need: Rethinking STEM Education with Prompt Engineering — arXiv, Dec 2024 (doi.org/10.48550/arXiv.2412.05023)
- Revolutionizing High School Physics Education: A Novel Dataset — Big Data and AI Conference, Dec 2023 (Springer)